

CPU Scheduler Simulation Report

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1 Introduction

This project simulates the behavior of classical CPU scheduling algorithms in an interactive web environment using Python and Dash. The goal is to analyze and visualize how different strategies affect CPU performance and process turnaround.

2 Implemented Algorithms

The following algorithms were implemented:

- First-Come, First-Served (FCFS)
- Shortest Job First (SJF)
- Priority Scheduling
- Round Robin (RR)
- Hybrid Priority + Round Robin

3 Process Representation

Each process is defined by:

- Process ID
- Arrival Time
- Burst Time
- Priority (if applicable)

4 User Interaction

Users can:

- Insert process information manually
- Generate random processes
- Read processes from a file (e.g., `processes.txt`)
- Select an algorithm via dropdown
- Compare algorithms based on turnaround time, waiting time, and CPU usage

5 Visualization Features

The app provides:

- Completion, turnaround, waiting, and start times per process
- Gantt chart
- CPU usage over time
- Global metrics (avg TAT, avg WT, CPU Utilization)
- Algorithm comparison chart

6 Performance Metrics

- Average Turnaround Time
- Average Waiting Time
- CPU Utilization

7 Testing

We tested the simulator with various configurations:

- Small and large process counts
- Different arrival/burst/priority values
- Edge cases (zero burst, negative arrival, etc.)

8 Conclusion

The scheduler successfully demonstrates how algorithmic choices affect process execution and system performance. The Dash interface provides an intuitive way to visualize complex scheduling behavior and compare strategies.